QUESTION 1

Which of the following decide the output of the ALU at any given time?

Answer is Wrong **Score: 0 / 1**

Options

Wrong

A. clock

Expected

B. registers

Expected

C. instruction

D. temperature

FEEDBACK

Feedback

The output of the Arithmetic Logic Unit (ALU) at any given time is determined by both the instruction being executed and the registers that provide the necessary data inputs. Here's how these elements influence the ALU's output:

Correct Options:

- Registers (b): Registers hold the operands (data) that the ALU uses for its operations. For example, in an addition operation, the ALU will use the values stored in the specified registers to compute the result. The content of these registers directly influences the result produced by the ALU.

- Instruction (c): The instruction, fetched from memory and interpreted by the CPU, specifies what operation the ALU should perform (e.g., addition, subtraction, logical operations). The instruction tells the ALU which registers to use as operands and what operation to carry out on those operands. Therefore, the specific operation determined by the instruction directly affects the ALU's output.

Incorrect Options:

- Clock (a): While the clock synchronizes all operations within the CPU, including those of the ALU, it does not directly determine the specific output of the ALU. The clock ensures that operations occur at the correct times but does not influence the data or the specific operation being performed.

- Temperature (d): Temperature can affect the physical performance and reliability of the hardware but does not play a role in determining the logical output of the ALU for any specific operation.

In summary, both the registers (providing data) and the instruction (specifying the operation) are crucial in deciding the output of the ALU at any given time.

QUESTION 2

Which of the following is true for a real-time clock in uC systems?

Not Attempted **Score: 0 / 1**

Options

A. fastest clock in the system

B. slowest clock in the system

Expected

C. can wake the system from deep sleep

D. system must always run at a faster speed than RTC

FEEDBACK

Feedback

Let's analyze each option to understand why (c) is correct and the others are not typically true for a real-time clock (RTC) in microcontroller systems:

- (a) Fastest clock in the system: This is incorrect. The RTC is usually a low-frequency clock designed for timekeeping purposes. It operates at a much slower rate than the main system clock to minimize power consumption and ensure accurate timekeeping over long periods.
- (b) Slowest clock in the system: This is more likely to be true, but it is not necessarily the defining characteristic of an RTC. The RTC is indeed one of the slower clocks in the system, but the key feature is its ability to keep accurate time rather than just being the slowest clock.

- (c) Can wake the system from deep sleep: This is correct. One of the primary functions of an RTC in a microcontroller system is to keep track of time even when the system is in a low-power or deep sleep mode. The RTC can generate interrupts or alarms that wake up the microcontroller to perform tasks at specific times or intervals.

- (d) System must always run at a faster speed than RTC: This is incorrect. The system clock typically runs at a higher frequency than the RTC when active, but this statement is misleading. The system does not "always" run; it can enter sleep modes where the main system clock is stopped or greatly reduced, while the RTC continues to run.

Given these explanations, the correct answer is:

(c) can wake the system from deep sleep.

QUESTION 3

A 10-bit ADC can digitize an analog value into ___ distinct levels.

Not Attempted **Score: 0 / 1**

Your answer

No answer given

Expected answer

Expected answer: `1024`

FEEDBACK

Feedback

An Analog-to-Digital Converter (ADC) converts an analog input signal into a digital output signal. The resolution of an ADC indicates how many distinct levels or steps it can use to represent the analog input. This is determined by the number of bits in the ADC.

- A 10-bit ADC has 10 bits for each conversion.

To find the number of distinct levels, you use the formula:

Number of distinct levels =

For a 10-bit ADC:

Number of distinct levels =

Therefore, a 10-bit ADC can digitize an analog value into 1024 distinct levels.

QUESTION 4

If you use an 8-bit ADC to digitize a signal in the range -5 to +5V, what is the minimum difference(approx) between two voltages that you can measure?

Not Attempted **Score: 0 / 1**

Options

A. 10 mV

B. 20 mV

C. 30 mV

Expected

D. 40 mV

FEEDBACK

Feedback

To determine the minimum voltage difference that can be measured with an 8-bit ADC for a given voltage range, follow these steps:

1. Determine the resolution: An 8-bit ADC can resolve distinct levels.

2. Calculate the voltage range per level:

– Given the range is from -5V to +5V, the total span is .

3. Calculate the voltage resolution:

– Divide the total voltage range by the number of levels:

4. Convert the resolution to millivolts (mV):

–

5. Select the minimum difference between two voltages:

– The closest option is (d) 40 mV, which is the closest practical approximation given the calculated resolution.

Conclusion:

Therefore, with an 8-bit ADC digitizing a signal in the range of -5V to +5V, the minimum difference between two voltages that can be measured is approximately 40 mV, making option (d) the correct answer.

QUESTION 5

What is the purpose of the sampling time parameter in ADC configuration?

Not Attempted **Score: 0 / 1**

Options

A. To set the maximum voltage that can be measured

Expected

B. To determine how long the ADC samples the input signal before conversion

C. To configure the clock speed of the microcontroller

D. To select the analog input channel to be converted

FEEDBACK

Feedback

Explanation:

- (a) To set the maximum voltage that can be measured: This is typically determined by the reference voltage provided to the ADC, not the sampling time.
- (b) To determine how long the ADC samples the input signal before conversion: This is correct. The sampling time determines the duration for which the ADC captures and holds the input voltage before converting it to a digital value. A longer sampling time can improve accuracy by allowing the ADC to settle to the input voltage.

- (c) To configure the clock speed of the microcontroller: The clock speed of the microcontroller is not directly related to the sampling time of the ADC.
- (d) To select the analog input channel to be converted: This is typically done through a separate channel selection parameter, not the sampling time.

Therefore, the correct answer is (b). It's essential for ensuring that the ADC captures an accurate representation of the analog signal before converting it into a digital value.

QUESTION 6

Which parameter of a PWM signal determines the average power delivered to a load?

Not Attempted **Score: 0 / 1**

Options

A. Frequency

Expected

B. Duty cycle

C. Amplitude

D. Phase

FEEDBACK

Feedback

Explanation:

- (a) Frequency: The frequency of a PWM signal determines how often the pulses occur within a given time period but does not directly determine the average power delivered to the load.
- (b) Duty cycle: The duty cycle of a PWM signal specifies the percentage of time the signal is high (ON) compared to the total period. The average power delivered to the load is proportional to the duty cycle. For example, a higher duty cycle means the signal is ON for a larger portion of the time, delivering more average power to the load.

- (c) Amplitude: In the context of PWM, the amplitude refers to the peak voltage of the signal, which affects the signal swing but does not determine the average power delivered to the load.
- (d) Phase: Phase refers to the timing relationship between multiple waveforms and is not directly related to determining the average power delivered to a load in a single PWM signal.

Therefore, the correct answer is (b) Duty cycle, as it directly controls the average power delivered to a load in PWM applications.

QUESTION 7

In PWM, what does a 50% duty cycle mean?

Not Attempted **Score: 0 / 1**

Options

Expected

A. The signal is high for 50% of the time and low for 50% of the time

B. The signal is high for 50% of the time and low for 25% of the time

C. The signal is high for 75% of the time and low for 25% of the time

D. The signal is high for 25% of the time and low for 75% of the time

FEEDBACK

Feedback

In Pulse Width Modulation (PWM), the duty cycle refers to the ratio of time the signal is ON (high) compared to the total period of the signal. Here's how a 50% duty cycle is defined:

- (a) The signal is high for 50% of the time and low for 50% of the time: This statement is correct for a 50% duty cycle. It means that the signal is ON (high) for half of the total period and OFF (low) for the other half of the period.
- (b) The signal is high for 50% of the time and low for 25% of the time: This describes a duty cycle of 66.7%, not 50%.
- (c) The signal is high for 75% of the time and low for 25% of the time: This describes a duty cycle of 75%, not 50%.

- (d) The signal is high for 25% of the time and low for 75% of the time: This describes a duty cycle of 25%, not 50%.

Therefore, the correct answer is (a). A 50% duty cycle in PWM means the signal spends an equal amount of time ON and OFF, effectively delivering half of the maximum possible power to the load.

QUESTION 8

Reference manual Link: Link

Which timer feature is used to measure the time interval between external events?

Not Attempted **Score: 0 / 1**

Options

A. PWM (Pulse Width Modulation)

Expected

B. Input capture

C. Output compare

D. Watchdog timer

FEEDBACK

Feedback

Explanation:

- (a) PWM (Pulse Width Modulation): PWM is used to generate signals with varying duty cycles, not specifically for measuring time intervals between external events.
- (b) Input capture: Input capture is a timer feature that captures the timer value when a specified external event (such as a rising or falling edge on an input pin) occurs. It is commonly used to measure time intervals between external events accurately.

- (c) Output compare: Output compare is used to generate events (e.g., output a signal or trigger an interrupt) when the timer value matches a predefined compare value. It is not primarily used for measuring time intervals between external events.
- (d) Watchdog timer: The watchdog timer is a separate timer used for system reset or recovery if the software hangs or crashes, not for measuring time intervals between external events.

Therefore, the correct answer is (b) Input capture, as it specifically allows the microcontroller to capture and measure the time interval between external events accurately using timer peripherals.

QUESTION 9

Which timer mode is used to generate a PWM signal?

Not Attempted **Score: 0 / 1**

Options

A. Input capture mode

Expected

B. Output compare mode

C. Timer overflow mode

D. Watchdog timer mode

FEEDBACK

Feedback

Explanation:

- (a) Input capture mode: Input capture mode captures the timer value when a specific event occurs on an external pin. It is used for measuring time intervals between external events, not for generating PWM signals.
- (b) Output compare mode: This is the correct answer. In output compare mode, the timer compares its value with a predefined compare value. When the timer value matches the compare value, it can toggle an output pin, generate an interrupt, or perform other actions. By adjusting the compare value dynamically, PWM signals with varying duty cycles can be generated.

- (c) Timer overflow mode: Timer overflow mode is when the timer counts up to its maximum value and then overflows back to zero. It is used for generating interrupts at fixed intervals, not directly for generating PWM signals.
- (d) Watchdog timer mode: The watchdog timer is a separate timer that resets the microcontroller if a certain task or operation is not completed within a specified time frame. It is not used for generating PWM signals.

Therefore, the correct and specific timer mode used to generate PWM signals in microcontroller applications is (b) Output compare mode. This mode allows for flexible control of PWM signals by adjusting the compare value to change the duty cycle dynamically.

QUESTION 10

An integrated circuit has a capacitance (C) of 50 pF and operates at a voltage (V) of 1.2 V. If the circuit operates at a clock frequency (f) of 100 MHz, calculate its power consumption.

Not Attempted **Score: 0 / 1**

Options

Expected

A. 7.2 mW

B. 12.0 mW

C. 18.0 mW

D. 24.0 mW

FEEDBACK

Feedback

Given:

- Capacitance, $C = 50 pF = 50 × 10^{-12} F$

- Voltage, $V = 1.2 V$

- Frequency, $f = 100~ MHz = 100 × 10^6 Hz$

Using the formula for dynamic power consumption (P):

$P = CV^2f$

Substitute the given values:

$P = (50 \times 10^{-12}) \times (1.2)^2 \times (100 \times 10^6)$

Calculate V^2:

$V^2 = 1.2^2 = 1.44$

Now, calculate P:

$P = 50 \times 10^{-12} \times 1.44 \times 100 \times 10^6$

$P = 7.2 \times 10^{-12} \times 10^8$

$P = 7.2 \times 10^{-4}$

$P = 7.2 \times 10^{-3} \text{ mW}$

$P = 7.2 \text{ mW}$

Therefore, the dynamic power consumption (P) of the integrated circuit is approximately **7.2 mW**.

QUESTION 11

Which of the following is a common application of PWM in microcontroller systems?

Not Attempted **Score: 0 / 1**

Options

A. Data transmission

B. Temperature sensing

Expected

C. Generating analog signals from digital outputs

D. Real-time clock implementation

FEEDBACK

Feedback

Explanation:

- (a) Data transmission: PWM is not typically used for data transmission in microcontroller systems. Communication protocols like UART, SPI, or I2C are more suitable for data transmission.
- (b) Temperature sensing: Temperature sensing involves reading analog signals from sensors, not generating PWM signals.
- (c) Generating analog signals from digital outputs: This is correct. PWM is widely used to generate analog-like signals from digital outputs by varying the duty cycle of the PWM signal. By averaging the signal over time, PWM can effectively control the output voltage or current, making it useful for applications like controlling motor speed, LED brightness, or DAC (Digital-to-Analog Converter) simulation.

- (d) Real-time clock implementation: Real-time clock (RTC) implementation involves keeping accurate time and waking up the microcontroller at specific intervals using a dedicated RTC module, not PWM.

Therefore, the correct and relevant answer in the context of PWM applications in microcontroller systems is (c).